Course: MENG 2020 Computer Programming	
Lab: Lab 03	
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Please refer to the lab manual posted on the blackboard and fill in the following table as your final lab report.

Please Notice: All the labs are individual work and submission. All academic misconduct behaviors will be penalized the maximum extent possible according to college policy. Thanks for your attention. (Reference Link: About Academic Misconduct)

Please finish all the ***lab report submission requirements*** in the following table. Add more rows if necessary.

Lab Results

Lab Procedures	**Submission Requirements**: Please submit ***screenshots*** showing your original Matlab code and the execution results for each step. ***Note***: If file submission is required, please attach your files separately on Blackboard.
1	

```
format bank
nthyear = [0:5:30]; L = 100000; r = 0.06; n = 30;
B = L*((1 + r/12)^(12*n) - (1 + r/12).^(12*nthyear)) ./ ((1 + r/12)^(12*n) - 1);

disp(['Chapter 3, Question 7:' newline]);
disp([nthyear; B])
pause; clc; clear;
```

```
Chapter 3, Question 7:

           0         5.00        10.00        15.00        20.00        25.00        30.00
   100000.00     93054.36     83685.72     71048.84     54003.59     31012.09            0
```

```
disp(['Chapter 3, Question 8:' newline]);
u = [-5.6, 11, -14];
length_u = sqrt(sum(u .* u));
disp(length_u);
pause; clc; clear;
```

```
Chapter 3, Question 8:

         18.66
```

```
disp(['Chapter 3, Question 9:' newline 'A)' newline ' U =']);
u = [4,13,-7];
disp(u);
pause;

disp(['Chapter 3, Question 9:' newline 'B)' newline ' Lu =']);
Lu = sqrt( sum(u.^2));
disp(Lu);
pause;

disp(['Chapter 3, Question 9:' newline 'C)' newline ' un =']);
un = (u/Lu);
disp(un);
pause;

disp(['Chapter 3, Question 9:' newline 'D)' newline ' un =']);
un = sqrt ( sum(un.^2));
disp(un);
pause; clc; clear;
```

```
Chapter 3, Question 9:
A)
 U =
          4.00         13.00         -7.00

Chapter 3, Question 9:
B)
 Lu =
         15.30

Chapter 3, Question 9:
C)
 un =
          0.26          0.85         -0.46

Chapter 3, Question 9:
D)
 un =
          1.00
```

```
disp(['Chapter 3, Question 10:']);
u1 = [3.2, -6.8, 9];
u2 = [-4, 2, 7];
theta = acosd(sum(u1 .* u2) / (sqrt(sum(u1 .* u1)) * sqrt(sum(u2 .* u2))));
disp(theta);
pause; clc; clear;
```

```
Chapter 3, Question 10:
         67.93
```

2

```
disp('Chapter 3, Question 20:');
a = [7, -4, 6];
b = [-4, 7, 5];
c = [5, -6, 8];

LHS = cross(a, cross(b,c));
disp("Left Hand Side:");
disp(LHS);
RHS = (b*dot(a,c))-(c*dot(a,b));
disp("Right Hand Side:");
disp(RHS);
identity = isequal(LHS,RHS);
disp("Should return with a 1 if the identity is equal to each other:")
disp(identity);
pause; clc; clear;
```

```
Chapter 3, Question 20:
Left Hand Side:
       -298.00        593.00        743.00

Right Hand Side:
       -298.00        593.00        743.00

Should return with a 1 if the identity is equal to each other:
   1
```

```
format long

disp(['Chapter 3, Question 22:' newline 'A)' newline 'n = ']);
n1 = 1:5;
sum5 = sum(1./n1.^2);
disp(sum5);
pause;

disp(['Chapter 3, Question 22:' newline 'B)' newline 'n = ']);
n2 = 1:50;
sum50 = sum(1./n2.^2);
disp(sum50);
pause;

disp(['Chapter 3, Question 22:' newline 'C)' newline 'n = ']);
n3 = 1:5000;
sum5000 = sum(1./n3.^2);
disp(sum5000);
pause;

exact = (pi^2)/6;

err = exact - [sum5, sum50, sum5000];
disp("The sum of 5000 is the closest.");
disp(err);
pause; clc; clear;
```

```
Chapter 3, Question 22:
A)
n =
   1.463611111111111

Chapter 3, Question 22:
B)
n =
   1.625132733621529

Chapter 3, Question 22:
C)
n =
   1.644734086846895

The sum of 5000 is the closest.
   0.181322955737115   0.019801333226697   0.000199980001331
```

```
format long
x = 2;

disp(['Chapter 3, Question 24:' newline 'A)' newline 'n = ']);
n1 = 0:5;
sum5 = sum(x.^n1 ./ factorial(n1));
disp(sum5);
pause;

disp(['Chapter 3, Question 24:' newline 'B)' newline 'n = ']);
n2 = 0:15;
sum15 = sum(x.^n2 ./ factorial(n2));
disp(sum15);
pause;

disp(['Chapter 3, Question 24:' newline 'C)' newline 'n = ']);
n3 = 0:25;
sum25 = sum(x.^n3 ./ factorial(n3));
disp(sum25);
pause;

exact = exp(2);

err = exact - [sum5, sum15, sum25];
disp("The sum of 25 is the closest.");
disp(err);
pause; clc; clear;
```

```
Chapter 3, Question 24:
A)
n =
   7.266666666666667

Chapter 3, Question 24:
B)
n =
   7.389056095384136

Chapter 3, Question 24:
C)
n =
   7.389056098930649

The sum of 25 is the closest.
   0.122389432263984   0.000000003546514   0.000000000000002
```

```
A = [5 -3 7; 1 0 -6; -4 8 9];
B = [3 2 -1; 6 8 -7; 4 4 0];
C = [-9 8 3; 1 7 -5; 3 3 6];

disp(['Chapter 3, Question 29:' newline 'A)']);
id_1 = A+B;
id_2 = B+A;

disp(id_1);
disp(id_2);
disp("Should return with a 1 if the identity is equal to each other:");
id_ans1 = isequal(id_1, id_2);
disp(id_ans1);
pause;

disp(['Chapter 3, Question 29:' newline 'B)']);
id_3 = A*(B*C);
id_4 = (A*B)*C;

disp(id_3);
disp(id_4);
disp("Should return with a 1 if the identity is equal to each other:");
id_ans2 = isequal(id_3, id_4);
disp(id_ans2);
pause;

disp(['Chapter 3, Question 29:' newline 'C)']);
id_5 = 5*(B+C);
id_6 = (5*B)+(5*C);

disp(id_5);
disp(id_6);
disp("Should return with a 1 if the identity is equal to each other:");
id_ans3 = isequal(id_5, id_6);
disp(id_ans3);
pause;

disp(['Chapter 3, Question 29:' newline 'D)']);
id_7 = (A+B)*C;
id_8 = (A*C)+(B*C);

disp(id_7);
disp(id_8);
disp("Should return with a 1 if the identity is equal to each other:");
id_ans4 = isequal(id_7, id_8);
disp(id_ans4);
pause; clc; clear;
```

```
Chapter 3, Question 29:
A)
     8    -1     6
     7     8   -13
     0    12     9

     8    -1     6
     7     8   -13
     0    12     9

Should return with a 1 if the identity is equal to each other:
   1
Chapter 3, Question 29:
B)
        -163         346         101
         164        -325          41
        -712        1064        -556

        -163         346         101
         164        -325          41
        -712        1064        -556

Should return with a 1 if the identity is equal to each other:
   1
Chapter 3, Question 29:
C)
   -30    50    10
    35    75   -60
    35    35    30

   -30    50    10
    35    75   -60
    35    35    30

Should return with a 1 if the identity is equal to each other:
   1
Chapter 3, Question 29:
D)
   -55    75    65
   -94    73   -97
    39   111    -6

   -55    75    65
   -94    73   -97
    39   111    -6

Should return with a 1 if the identity is equal to each other:
   1
```

```
format bank
disp(['Chapter 3, Question 33:']);
%eq1 -> -2x+5y+7z = -17.5
%eq2 -> 3x-6y+2z = 40.6
%eq3 -> 9x-3y+8z = 56.2
LHS = [-2 5 7; 3 -6 2; 9 -3 8];
RHS = [-17.5; 40.6; 56.2];
equ = LHS \ RHS;
disp(equ);
pause; clc; clear;
```

```
Chapter 3, Question 33:
          3.20
         -4.60
          1.70
```

```
disp(['Chapter 3, Question 37:']);
V1 = 28; V2 = 36; V3 = 42;
R1 = 16; R2 = 10; R3 = 6;
R4 = 12; R5 = 8;  R6 = 14;
R7 = 4; R8 = 5;

% Matrix A %
A = [-(R1+R3),R3,0,0;
    R3,-(R2+R3+R4+R6),R4,R6;
    0,R4,-(R4+R5+R7),R7;
    0,R6,R7,-(R6+R7+R8)];
% Matrix B %
B = [-V1;0;V2;-V3];

% Calculation's %
x = A\B;
i1 = x(1); i2 = x(2); i3 = x(3); i4 = x(4);

% Current through each resistor
I_R1 = i1;
I_R2 = i2;
I_R3 = i1 - i2;
I_R4 = i2 - i3;
I_R5 = i3;
I_R6 = i4 - i2;
I_R7 = i4 - i3;
I_R8 = i4;

fprintf('Mesh currents:\ni1=%.4f, i2=%.4f, i3=%.4f, i4=%.4f A\n', i1,i2,i3,i4);
fprintf('R1=%.4f R2=%.4f R3=%.4f R4=%.4f\n', I_R1,I_R2,I_R3,I_R4);
fprintf('R5=%.4f R6=%.4f R7=%.4f R8=%.4f\n\n', I_R5,I_R6,I_R7,I_R8);

% To verify if its correct it should output the same values as B %
QED = A*x;
disp("QED =");
disp(QED);
```

```
Chapter 3, Question 37:
Mesh currents:
i1=1.4565, i2=0.6737, i3=-0.8141, i4=2.0946 A
R1=1.4565 R2=0.6737 R3=0.7828 R4=1.4877
R5=-0.8141 R6=1.4209 R7=2.9086 R8=2.0946

QED =
        -28.00
          0.00
         36.00
        -42.00
```

After Lab Practice Questions

```
h1:[10 14 8 6]
h2:[6 4 12 2]
m:h1'+h2
 h:h1-h2
 r:h1.*h2
 s:h1.*h2'
s1:dot[h1,h2]
s2:dot[h1',h2]
h1:[100:100:400]
p:h1'+h1'''
q:h1'+h1'
[t,u]:max[h1]
h1:[10:20:60]
h2:'This is for fun?'
v:length[h1]
vv:length[h2]
```

Variable Name	Variable Value
M	16 18 20 8
N	4 8 2 0 6 10 4 2 -2 2 -4 -6 8 12 6 4
P	200 400 600 800
Q	200 400 600 800
R	60 84 48 36 40 56 32 24 120 168 96 72 20 28 16 12
S	60 56 96 12
S1	224
S2	224
T	400
U	4
V	16
W	3